Real-Time Data Storage, Prediction Output & Power BI Dashboard Visualization

1. Objective

The objective of this milestone is to:

- Store real-time sensor data in a **structured format**
- Generate and store **spoilage prediction outputs**
- Develop an **interactive Power BI dashboard** to visualize:
    - Storage conditions (temperature, humidity, gas)
    - Trends over time
    - Spoilage alerts and predictions
    - Live monitoring insights

2. System Overview

In this phase, the system collects real-time environmental data from the ESP32-based cold storage monitoring setup. The collected data is:

1. Sent to a cloud platform (ThingSpeak / API)
2. Stored in a structured dataset (Excel)
3. Used for prediction (Safe / At Risk)
4. Visualized using Power BI

3. Data Collection & Storage

3.1 Data Source

- Sensor: **DHT22 (Temperature & Humidity)**
- Gas sensor readings (as per your code logic)
- Door status (open/close count)

3.2 Data Transmission

- ESP32 sends data via WiFi using HTTP requests
- Data is uploaded to cloud (ThingSpeak API)

3.3 Structured Storage Format

The real-time data is stored in an **Excel file (live_predictions3.xls)** with the following fields:

Column Name	Description
Timestamp	Date & time of reading
Temperature	Measured temperature (°C)
Humidity	Measured humidity (%)
Gas	Gas sensor value
Door_Count	Number of door openings
Spoiled_Food	Actual spoilage (Yes/No)
Prediction	Model output (Safe / At Risk)

✓ This structured format ensures:

- Easy analysis
- Compatibility with Power BI
- Efficient real-time updates

4. Prediction System

4.1 Logic

The system predicts spoilage based on:

- High temperature
- High humidity
- Abnormal gas levels

4.2 Output Categories

- **Safe** → Normal storage conditions
- **At Risk** → Potential spoilage detected

4.3 Storage of Predictions

- Predictions are stored alongside real-time data in Excel
- Enables historical analysis and trend tracking

5. Power BI Dashboard Development

5.1 Data Import

- Excel file (live_predictions3.xls) is connected to Power BI
- Data is set to **auto-refresh** for near real-time updates

6. Dashboard Components (Based on Your Image)

6.1 KPI Cards (Top Section)

Displays key metrics:

- **Avg Temperature** → 5.13°C
- **Spoilage %** → 23.76%
- **Total Door Openings** → 17
- **Avg Humidity** → 86.70%
- **Avg Gas** → 2.08K

✓ Purpose:

- Quick overview of storage condition health

6.2 Time-Series Graphs

1. **Temperature vs Timestamp**
    - Shows fluctuations over time
    - Helps detect abnormal spikes
2. **Gas vs Timestamp**
    - Tracks gas accumulation trends
    - Includes threshold line for alerts

✓ Purpose:

- Identify trends and anomalies

6.3 Donut Charts

1. **Spoiled Food Distribution**
    - YES vs NO
    - Example: 23.76% spoiled
2. **Prediction Distribution**
    - Safe vs At Risk
    - Example:
        - 74.26% Safe
        - 25.74% At Risk

✓ Purpose:

- Visual representation of spoilage and risk levels

6.4 Tooltip / Live Data Insight

- Hover feature shows:
    - Exact timestamp
    - Gas value
- Enables detailed inspection of specific data points

7. Real-Time Data Handling in Power BI

Steps Implemented:

1. Load Excel dataset
2. Use **Power Query** for cleaning
3. Create calculated measures:
    - Average values
    - Spoilage percentage
4. Enable **Auto Refresh**
5. Design interactive visuals

8. Key Features of Dashboard

✓ Real-time monitoring
✓ Interactive filtering
✓ Trend analysis
✓ Spoilage detection
✓ Visual alerts
✓ User-friendly layout

9. Benefits of This Milestone

- Enables **early spoilage detection**
- Improves **cold storage efficiency**
- Reduces **food wastage**
- Provides **data-driven decision making**
- Supports **predictive maintenance**

10. Conclusion

In Milestone 4, a complete pipeline was successfully developed to:

- Capture real-time sensor data
- Store it in a structured format
- Generate spoilage predictions
- Visualize insights using an interactive Power BI dashboard

This milestone transforms raw IoT data into **meaningful, actionable insights**, making the system intelligent and efficient for real-world cold storage monitoring.